OpenPI Comet — Paper Manager pack (smoke run)

Paper: Openpi Comet: Competition Solution For 2025 BEHAVIOR Challenge

arXiv: https://arxiv.org/abs/2512.10071 · PDF: https://arxiv.org/pdf/2512.10071.pdf

Code: https://github.com/mli0603/openpi-comet · HF: sunshk/openpi_comet

Schedule (Sep 2 smoke): Stakeholder=1 (Kanta), Reviewer=2, Empiricist=4, Archaeologist=2, Visionary=3

Sources: Paper Summarizer + Senior Research (Ablation Experimenter) + Ablation inventory

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1. Paper summary

Team Comet adapts public **$\pi_{0.5}$ (OpenPI)** to **50** long-horizon BEHAVIOR-1K household tasks with a data mix (teleop + planner + offline RL), multi-task pre-train coverage (pt1→pt50), **Rejection Sampling Fine-Tuning (RFT)**, and **receding-horizon** inference — **no** modular skill chaining.

- **Contest:** 2nd place — test Q **0.2514**, **22/50** tasks (winner 0.2599).

- **Post-challenge** (refined balancing): val Q **0.3453**, SR **0.1500** (above winner's val Q 0.2605).

- **Story:** Public VLA + data flywheel is competitive; hard half of the suite and a **0.345** → **0.611** oracle gap remain.

Method in one line: visual (head+wrist RGB) + language + proprio → transformer action expert → continuous control; pre-train ~1.5k hours; RFT N=3 (~8500 rollouts/round → **1469** kept); serve with receding horizon, action horizon **32**, absolute joints **30 Hz**, state on.

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2. Important model / keywords

Kind	Terms
Backbone	$\pi_{0.5}$, OpenPI, VLA, flow/diffusion action expert
Benchmark	BEHAVIOR-1K, BDDL, Q-score, NeurIPS 2025 BEHAVIOR Challenge

Training	SFT, multi-task pre-train (pt1/7/10/50), RFT, rejection sampling
Inference	Receding horizon, temporal ensemble, action horizon H=32
Data / control	Absolute vs delta joints, 30 Hz vs 15 Hz, proprio state, RGB vs depth vs point cloud
Metrics	Success rate (SR), Q-score, theoretical-best oracle (0.611)
Stack	JAX, OmniGibson, H200 batch 64/device

Default ablation baseline (Tables 3–4): Receding Horizon, H=32, RGB 224×224, absolute joints, 30 Hz, state on — task `turning_on_radio`.

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3. Role-ready bullets (for slides / talk)

Stakeholder (18+3) — Kanta on this schedule

- **Motivation:** Minute-scale household skill chaining fails; BEHAVIOR is the public scoreboard.

- **Problem:** 50 full BDDL activities; rank by test Q; bet one end-to-end public VLA can compete without a skill graph.

- **Method:** $\pi_{0.5}$ + **motion-planner/offline-RL data** + coverage ladder + 3-round RFT + **receding-horizon** control (H=32), abs/30Hz/state/RGB — no modular skill planner at inference.

- **Findings:** Contest 2nd (0.2514); post-challenge val Q 0.3453; wrong controller/Hz/frame → ~0%; high-res doubles probe SR; oracle 0.611 = headroom.

- **Takeaway:** Public VLA + flywheel works; gap is post-train efficiency + hard tasks.

Scientific Reviewer (8+3)

- **Strengths:** Reusable recipe; coverage ablation; specified RFT; inference/data ablations; honest headroom; code out.

- **Weaknesses:** Q-score undefined; ablations mostly on easy `turning_on_radio`; post-challenge missing test Q; oracle easy to misread; no error bars; no modular baseline.

- **Actionable fixes:** Define Q; hard-task ablation panel; post-challenge test Q; mean±std; publish balance weights; hierarchical baseline; held-out RFT; turn oracle into a router.

Empiricist (11+3)

- **H1:** Under RH + abs joints + proprio + RGB, **H=32 ≫ H=8** on `turning_on_radio` (paper 0.30 vs 0.00).

- **Implement:** clone openpi-comet + BEHAVIOR-1K; warm-start HF `pi05-b1kpt12-cs32`; two short SFT configs (~2k steps); `serve_b1k` receding_horizon; eval 5–10 instances.

- **Success:** ordinal gap H32−H8 ≥ +0.15 (or H32>0, H8≈0). Absolute 0.30 / full Q **out of scope** for smoke.

- **Lighter alt:** fix H32 ckpt; eval-only control-mode (RH vs temporal ensemble).

Archaeologist (11+3)

- **Priors that shaped it:** π_0 (Black et al.); RT-1/RT-2; Octo/OpenVLA; BEHAVIOR-1K (Li et al. 2024); skill-chaining lit (rejected); LLM-style RFT transplanted; PiRL/RLinf (why online RL is hard here).

- **Lineage:** Adaptation study, not a new VLA architecture.

- **Subsequent:** Peer-reviewed follow-ups unclear; team released repo, post-challenge ckpts, HF RFT set (`comet-1.5k`).

Visionary (11+3) — content only (no Discord/Drive)

- → **2026 BEHAVIOR:** Edge = sample-efficient post-train (DAgger/experts/rewards), skill curriculum, eval-time perception fidelity — not opaque longer pretrain alone.

- **Follow-ups:** Privileged DAgger; checkpoint router; hard-task curricula; resolution/compute Pareto; sim-to-real of the *recipe*.

- **Don't over-claim:** Not a new foundation model; not real-robot deployment; household agent not solved.

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4. Empiricist run pack (if you are Empiricist)

Env setup (outline)

Full files: `/workspace/openpi-comet-smoke/env_outline/`

(`environment.yml`, `setup_mamba.sh`, sbatch scripts)

```
# Login node sketch (upstream prefers uv; hybrid mamba OK on Slurm)mamba env create -f env_outline/en
```

Train (smoke E01 — H=32 vs H=8)

```
export XLA_PYTHON_CLIENT_MEM_FRACTION=0.9# Configs: pi05_b1k-radio_h32 / pi05_b1k-radio_h8 — action_h
```

sbatch: `sbatch_smoke_train_h32.sbatch`, `sbatch_smoke_train_h8.sbatch`

Eval

```
# Job A — policy serveruv run scripts/serve_b1k.py \  --task_name=turning_on_radio \  --control_mode=
```

sbatch: `sbatch_smoke_eval.sbatch`

Estimated GPU time (E01 smoke)

Stage	Budget
SFT H=32, ~2k steps	~2–6 GPU-h
SFT H=8, same	~2–6 GPU-h
Eval 5–10 instances ×2 ckpts	~10–40 GPU-h wall (sim-bound)
Total	**~15–50 GPU-h**
RAM	LoRA **>22.5 GB** or full FT **>70 GB** — prefer **1×A100-80 / H100 / H200**

Prioritized smokes (from Senior Research)

Priority	ID	Experiment
P0	E01	Horizon H=8 vs 32 on `turning_on_radio`
P0	E02	Control mode RH vs temporal ensemble (eval-only)
P0	E03	Resolution 224 → 720/480
P0	E08	Single-task SFT from released HF ckpt (prerequisite)
Not smoke	E11–E14	Full pt50, full RFT, 50-task Q

Full table: `/workspace/openpi-comet-smoke/DELIVERABLE.md`

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5. Slide Maker brief (timing + visual style)

Role	Talk	Q&A	Visual bias
Stakeholder	18 min	3	System diagram (Fig1), leaderboard, coverage ladder
Reviewer	8 min	3	Strength/weakness cards; one ablation critique slide

Empiricist	11 min	3	H1 diagram; horizon table; env/run one-pager
Archaeologist	11 min	3	Citation timeline / lineage map
Visionary	11 min	3	BEHAVIOR 2026 bridge; follow-up map

Style: Prefer **images as the slide** (model figure, tables as figures); put bullet talking points in **speaker notes**. Avoid repeating Stakeholder background in later roles.

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6. Deep dive: Motion planning & long-horizon control (priority topic)

This is the paper's real engineering story: **not** a classical online motion planner at test time, but (1) **motion-planner / offline-RL data** in pre-training, plus (2) **receding-horizon action-chunk control** at inference, with carefully chosen horizon / action representation / rate / state.

6.1 What "planning" means in OpenPI Comet

Layer	What they do	What they deliberately *don't* do
Data	Mix human teleop with **~3.6K motion-planner demos + offline RL rollouts** (~0.4K hours)	Rely only on noisy human demos
Architecture	Single end-to-end **$\pi_{0.5}$** VLA (vision + language + proprio → continuous actions)	Modular skill graph / separate local policies + explicit skill chaining at deploy
Inference	**Receding Horizon**: predict an action segment of length **H**, execute it, then **re-plan** from new observations	Temporal Ensemble / Receding Temporal (open-loop smoothing → ~0% SR)
Training	Coverage ladder pt1→pt50 + RFT flywheel	Online RL in OmniGibson (too slow; split GPU needs)

Paper quote (paraphrased core claim): Temporal Ensemble / Receding Temporal fail closed-loop stability (~0 SR). Receding Horizon "executes all the predicted action segment and performs re-planning after finishing manipulation," and is necessary for long-horizon manipulation because smoothing open-loop predictions accumulates error.

6.2 Motion-planner data (why it matters)

Beyond the official **10k** teleop demos (>1,200 h):

- Extra **~3.6K** trajectories = **motion-planner demonstrations + offline RL rollouts**.

- **Planner demos:** precise, **low-noise** manipulation sequences (cleaner state–action pairs for contact-rich moves).

- **Offline RL rollouts:** broader behavioral variability (coverage the teleop set lacks).

- Together they “substantially enrich state–action coverage beyond human demonstrations.”

Stakeholder angle: Comet’s “planning” advantage starts in the **dataset**, not in a runtime planner module. The VLA *imitates* high-quality planned / optimized trajectories, then executes with closed-loop chunking.

Reviewer angle: Planner + RL stacks are **underspecified** (which planner? cost function? what offline RL algorithm?). Hard to reproduce the data mix exactly from the paper alone — code/HF help, but isolation of planner-data ablation vs teleop-only is not in Tables 3–4.

6.3 Why they reject classical skill-chaining planners

Intro argument:

- Long-horizon household tasks need **orchestrated interdependent behaviors**; compounding error + shifting state distributions kill open-loop / short-horizon VLAs.

- Common fix: decompose into subtasks + local policies — but that **doesn’t solve skill chaining** (reliable transitions between skills).

- Many chaining methods need online adaptation or modular stacks **incompatible** with large-scale offline end-to-end VLA training.

- Comet’s bet: push one public VLA with **data + training + inference design**, no skill graph at inference.

Archaeologist hooks: Konidaris skill chaining; SCAR; DiffSkill / hierarchical methods — cited as the rejected alternative lineage.

6.4 Inference-time control = the “online planner”

Default recipe that works on `turning_on_radio` ablations:

1. **Control mode = Receding Horizon** (SR **0.25** vs Temporal Ensemble **0.00** / Receding Temporal **0.00**)

2. **Action horizon H = 32** (non-monotonic: 8 → 0.00, 16 → 0.10, 50 → 0.25, **32 → 0.30**)

3. **Absolute joint** actions (delta → 0.00)

4. **30 Hz** sampling (15 Hz → 0.00)

5. **Proprioceptive state on** (off → 0.00)

6. RGB OK; high-res Head720/Wrist480 doubles SR (0.30 → 0.60); depth hurts; point cloud ≈ RGB but costly

Intuition to say out loud:

- **Too short H:** myopic; can't anticipate multi-stage behaviors.

- **Too long H:** conflicting temporal dependencies; short-term control becomes unreliable.

- **H=32** ≈ sweet spot between "look ahead" and "stay corrigible via replan."

- **Absolute joints + proprio + 30 Hz** = observability + temporal precision for closed-loop optimization.

- **Manip:Nav = 2:1 reweight does nothing** (0.30=0.30): navigation dominates frames (`move to` ~33%), but **resampling ≠ fixing** heterogeneous long-horizon dynamics.

6.5 Navigation vs manipulation (mobile manip "planning")

- Tasks need **coordinated navigation + precise manipulation** (multi-room, containers, multi-object).

- Skill imbalance: `move to` 33.3%, `pick up from` 24.4%, `place in` 8.8%, long tail 11.5%.

- Hard tasks (e.g. 48/49): long trajectories, 5–12+ skills, frequent **nav↔manip** switches = temporal credit assignment stress test.

- Easy bring-up tasks: avg length **<250 frames** — used to validate control stack before scaling.

Empiricist implication: Smoke on `turning_on_radio` tests the **control/horizon stack**, not full mobile multi-room planning. Say that explicitly in talk.

6.6 RFT as iterative "plan repair" offline

Algorithm 1 sketch:

1. Start from human demos + pretrained π.

2. Perturb initial pose → roll out → keep **BDDL successes only** → merge → retrain.

3. N=3 rounds; ~8500 traj/round → **1469** kept after dedup + task balancing.

4. Reject online RL: sim wall-clock 1h–1d/task; RT-core vs Tensor-core GPU split.

This is an **offline data flywheel** that improves robustness under pose disturbance — complementary to motion-planner demos (precision) and RH inference (closed-loop).

6.7 Talking points by role (motion-planning lens)

Stakeholder: "We didn't ship a skill planner. We taught $\pi_{0.5}$ on planner-quality trajectories, then ran it with receding-horizon re-planning. Wrong controller or wrong H → zero success."

Reviewer: “Strong inference ablations, but planner/RL data contribution never isolated; almost all control ablations on one easy task; ‘re-planning after finishing manipulation’ wording needs precise chunk-execute semantics vs classic MPC.”

Empiricist: Primary smoke = **H=32 vs H=8 under RH** (and optional control-mode eval-only). That *is* the motion-control claim you can falsify in a week of GPU.

Archaeologist: Place Comet between (a) classical / learned skill-chaining planners and (b) short-horizon VLAs — adaptation study that imports LLM-style RFT and MPC-like chunking into OpenPI.

Visionary → 2026 BEHAVIOR: Next wins likely from **structured long-horizon reasoning** (authors’ own conclusion), privileged DAgger / denser rewards, curricula over nav–manip imbalance, and maybe a **router over checkpoints** (oracle union Q 0.611) — not just longer open-loop chunks.

6.8 One diagram to put on a slide

```
[Human teleop] + [Motion planner demos] + [Offline RL]                          |                    ▼
```

Quick paths on box

Artifact	Path
This pack	`/workspace/openpi-comet-smoke/KANTA_PACK.md`
Senior Research deliverable	`/workspace/openpi-comet-smoke/DELIVERABLE.md`
Ablation inventory	`/workspace/papers/2512.10071-ablations.md`
PDF	`/workspace/openpi-comet-smoke/paper.pdf` or `/workspace/papers/2512.10071.pdf`
Env / sbatch	`/workspace/openpi-comet-smoke/env_outline/`